S.Y.BSc. 200109u- I

Semester IV Examination March 2023

VCD: 290328

All Questions are compulsory.

Figures to the right indicate full marks.

Time: 3 hours **Total marks: 100**

Q.1. a) Fill in the blanks. **5 Marks**

1. Archaea and Bacteria are grouped as _________ . (Prokaryotes, Eukaryotes, Vertebrates,Invertebrates)
2. __________ of genes is the process of flowing of genes into a population from nearby populations. (Mutation, Migration, Inbreeding, Deletion)
3. ______ of larger segment of chromosome proves to be fatal.(Deletion, Duplication, Inversion, Rotation)
4. ----------- means moving from general to specific approach to a problem under investigation. (induction, deduction, hypothesis,Abstract)
5. In an ----------the use of words should be 200-300 (abstract, thesis, bibliography, Analysis)

b) Match the column **5 Marks**

1. Oparin	a. Father of Paleontology
2. Leonardo da Vinci	b. Macroevolution
3. Stasis	c. Bottom to up
4. Induction	d. SPSS
5. Computer	e. Coacervates

c) Write true or false **5 Marks**

1. SBB stands for state biodiversity board.
2. Mass extinction is a process of rapid decrease in the amount of life on earth.
3. Bibliography is to express gratitude to all the people who have helped with the project.
4. In natural populations mating occurs as a selective process .
5. Protoplasm of all living organisms is different.

d) Write one sentence answer. **5 Marks**

1. What is Paleontology?
2. What is Inversion?
3. State the theory of Lamarckism.

4. Define research.
5. What is referred to as Mega-Evolution?

Q.2. Answer the following. (Any two) **20 Marks**

1. Discuss evidences in favor of organic evolution by giving examples of geographical distribution.
2. Discuss brief account of Origin of eukaryotic cell.
3. Explain Darwinism and Neo-Darwinism.
4. Describe Neutral theory of molecular evolution.

Q.3. Answer the following (Any two). **10 Marks**

1. Describe the different types of speciation.
2. Discuss the post-zygotic barriers which lead to reproductive isolation.
3. State Hardy Weinberg's law of equilibrium and discuss its salient features.
4. Describe the sources of genetic variation in natural populations.

Q.4. Answer the following (Any two). **20 Marks**

1. Explain different types of research.
2. Describe Steps in scientific method.
3. Give a detailed account on the Role of computers in data analysis.
4. Explain ethics in animal research.

Q.5. Write short notes. (Any four). **20 Marks**

1. Convergent and divergent evolution
2. Modern Synthetic theory
3. Miller-Urey experiment
4. Bibliography
5. Directional evolution in peppered moth
6. Characteristics of critical thinking
7. Genetic polymorphism
8. Haldane and Oparin theory